



Original Article

Enhanced piezoelectric properties and temperature stability of Bi₄Ti₃O₁₂-based Aurivillius ceramics via W/Nb substitutionXinchun Xie^a, Tianzi Wang^{a,b}, Zhiyong Zhou^{a,*}, Guofeng Cheng^c, Ruihong Liang^a, Xianlin Dong^{a,d,*}^a Key Laboratory of Inorganic Functional Materials and Devices, Shanghai Institute of Ceramics, Chinese Academy of Sciences, 588 Heshuo Road, Jiading, Shanghai 201899, People's Republic of China^b Suzhou Changfeng Avionics Co., Ltd, Suzhou 215151, People's Republic of China^c Analysis & Testing Center for Inorganic Materials, Shanghai Institute of Ceramics, Chinese Academy of Sciences, 588 Heshuo Road, Jiading, Shanghai, 201899, People's Republic of China^d State Key Laboratory of High Performance Ceramics and Superfine Microstructure, Shanghai Institute of Ceramics, Chinese Academy of Sciences, 1295 Dingxi Road, Shanghai 200050, People's Republic of China

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ABSTRACT

Bi₄Ti₃O₁₂ (BIT), a typical Aurivillius ceramics with high Curie temperature ($T_c \sim 675^\circ\text{C}$), has great potential for high temperature applications. This work provides an effective method of inducing structure distortion, relieving the tetragonal strain of the TiO₆ octahedron and decreasing the concentration of oxygen vacancies to improve the piezoelectricity and temperature stability of BIT ceramics. Bi₄Ti_{2.98}W_{0.01}Nb_{0.01}O₁₂ possesses an optimum piezoelectric coefficient (d_{33}) of 32 pC/N, a high T_c of 655 °C and a large resistivity of $3 \times 10^6 \Omega\cdot\text{cm}$ at 500 °C. The maximum d_{33} reported here is approximately quadruple than that of pure BIT (7 pC/N). Moreover, the d_{33} of W/Nb co-doped BIT and the *in-situ* temperature stability of the compression-mode sensor present a highly stable characteristic in the range of 25–600 °C. These results imply that W/Nb-modified BIT ceramics is a promising candidate for application at high temperatures of up to 600 °C.

1. Introduction

Aurivillius ferroelectrics (bismuth layer structure ferroelectrics, BLSFs) have attracted much attention owing to their high Curie temperature (T_c), good temperature stability and excellent fatigue-resistant properties [1–4]. Consequently, BLSFs ceramics display excellent potential for applications in high temperature piezoelectric sensor systems and nonvolatile ferroelectric random-access memories (Nv RAMs) [5,6]. The general chemical formula of BLSFs is $(\text{Bi}_2\text{O}_2)^{2+}(\text{A}_{n-1}\text{B}_n\text{O}_{3n+1})^{2-}$, and its structure is an intergrowth of $(\text{A}_{n-1}\text{B}_n\text{O}_{3n+1})^{2-}$ units and $(\text{Bi}_2\text{O}_2)^{2+}$ slabs along the *c*-axis, where the A sites are larger cations (Na^+ , K^+ , Ca^{2+} , Sr^{2+} , Bi^{3+} etc.) and the B sites are smaller cations (Ti^{4+} , Nb^{5+} , V^{5+} , Mo^{6+} , W^{6+} etc.), with *n* ($1 \leq n \leq 6$) the number of oxygen octahedra in the pseudo-perovskite layers.

Bi₄Ti₃O₁₂ (BIT, *n* = 3), the prototype of BLSFs, is identified as a desirable candidate for application in the high temperature environment because of its high T_c ($\sim 675^\circ\text{C}$) and large ferroelectric polarization [7]. The spontaneous polarization (P_s) of BIT single crystals along the *a*-

axis can be as high as approximately $50 \mu\text{C}/\text{cm}^2$ [8]. However, the piezoelectric coefficient (d_{33}) of BIT is relatively low ($\sim 8 \text{ pC}/\text{N}$) due to its large coercive field (E_c) [9,10]. Furthermore, BIT presents typical *p*-type electrical conduction owing to Bi evaporation during sintering, which results in its very high electrical conductivity [11,12]. Therefore, the low piezoelectric activity and high electrical conductivity represent the dominant bottlenecks restricting the application of BIT ceramics. It's has been demonstrated that B site co-doped BIT-based ceramics by high valence cations substitution is an effective approach to enhance the electrical properties and decrease the concentration of defects. Hou et al. reported that a larger piezoelectric coefficient (d_{33}) of 26 pC/N could be obtained by Nb/Ta modification at the B-site for BIT ceramics, owing to the TiO₆ deformation and decrease of oxygen vacancies [13]. The DC resistivity of BIT was improved by three orders of magnitude after doping with W/Ni and a relatively large d_{33} of 18.5 pC/N has been achieved [14]. Chen et al. found that W/Cr substitution for the B-site of BIT ceramics can lead to the formation of defect dipoles, which further contribute to a large P_r of $17 \mu\text{C}/\text{cm}^2$ and a high d_{33} of 28 pC/N [15].

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For W/Mn co-substituted for BIT ceramics, a large piezoelectric activity ($d_{33} = 21$ pC/N) and a lower dielectric loss ($\tan\delta = 0.003$) were also reported, due to reduction of oxygen vacancies and the effect of Mn_{Ti} - V_O defect dipoles [16].

Though much progress has been made in recent years to improve the piezoelectric properties of BIT ceramics, the piezoelectric properties and temperature stability of BIT-based ceramics are still not sufficient to ensure excellent operational reliability at elevated temperatures. Therefore, it is still of vital importance to further enhance the piezoelectric properties of $\text{Bi}_4\text{Ti}_3\text{O}_{12}$ ceramics and its temperature stability. More importantly, high-temperature resistivity characteristics, as key factors for high-temperature application, need to be systematically investigated. Unfortunately, up to now only few reports are concerned with the reliability evaluation of the designed sensor fabricated with BIT-based ceramics at high temperature, which is of equal importance.

Based on the *p*-type conduction mechanism of many BLSF ceramics, it can be expected that the resistivity can be increased by donor doping and that good piezoelectric properties can be achieved via sufficient polarization. For example, $\text{Na}_{0.5}\text{Bi}_{4.5}\text{Ti}_4\text{O}_{15}$ -based ceramics with B-site substitution show excellent piezoelectric properties ($d_{33} = 28$ pC/N) with W/Nb co-doping [17]. Therefore, donor doping is expected to enhance electrical properties of BIT ceramics and further investigation of B site W/Nb-modified BIT-based ceramics is also of interest. In this work, we report significantly enhanced piezoelectric properties and temperature stability of BIT ceramics with B-site substitution using W/Nb ions by a solid solution reaction method. To clarifying the effects of W/Nb substitution on BIT ceramics, their structure characteristics, dielectric properties, DC resistivity and the temperature stability of piezoelectric properties were investigated in detail. Excellent charge sensitivity of the compression-mode sensor fabricated with W/Nb-modified BIT ceramics has been obtained at high temperatures.

2. Experimental procedure

Samples with compositions of $\text{Bi}_4\text{Ti}_{3-2x}\text{W}_x\text{Nb}_x\text{O}_{12}$ (BTWN-100x, $x = 0.00, 0.005, 0.010, 0.020$ and 0.030 , abbreviated as BIT, BTWN-0.5, BTWN-1, BTWN-2 and BTWN-3, respectively) were prepared by the conventional solid-state reaction method. All of the raw materials were Reagent-grade metal oxides, Bi_2O_3 (99.90%), TiO_2 (99.22%), WO_3 (99.00%) and Nb_2O_5 (99.90%). After weighing the raw materials, the powders were planetary-ball-milled for 4 h with anhydrous ethanol. The slurry was dried at 80°C and calcined at 850°C for 2 h. The calcined powders were reground under the same conditions. Then, the polyvinyl alcohol was used as a binder for granulation. Finally, the powders were pressed into a disk, approximately 13 mm in diameter and approximately 2 mm in thickness. The binder was burned out at 750°C , and then the green pellets were sintered at $1080\text{--}1140^\circ\text{C}$ for 2 h. For electrical characterizations, a platinum paste was coated onto the surfaces of the ceramic samples, then heated at 850°C for 30 min. The samples were polarized under a DC field of 6–10 kV/mm for 20 min.

The phase structure and microstructure of $\text{Bi}_4\text{Ti}_{3-2x}\text{W}_x\text{Nb}_x\text{O}_{12}$ ceramics were analyzed by Bruker D8 ADVANCE X-ray diffractometer, $\text{CuK}\alpha$ radiation (40 kV, 40 mA) and scanning electron microscopy (SEM; Hitachi, Tokyo, Japan), respectively. Raman spectra were investigated using a Jobin-Yvon micro-Raman spectrometer (LabRAW HR 800 UV, Paris, France). The DC electrical resistivity was measured by the means of a high-resistance measurement system (Model HRMS-900, Partulab, Wuhan, China) in the range of 200–600 $^\circ\text{C}$. The dielectric properties and piezoelectric constants d_{33} were determined using an LCR meter (Model E4980 A; Keysight, USA) and a d_{33} meter (Model ZJ-3; Institute of Acoustics, Chinese Academy of Sciences, Shanghai, China), respectively. Thermal depolarization tests were carried out by annealing the polarized samples at different temperatures for 4 h, then the d_{33} were measured repeatedly while cooling the samples to 25°C . The process was repeated at different temperatures. The charge sensitivity of the compression-mode accelerometer was measured using a

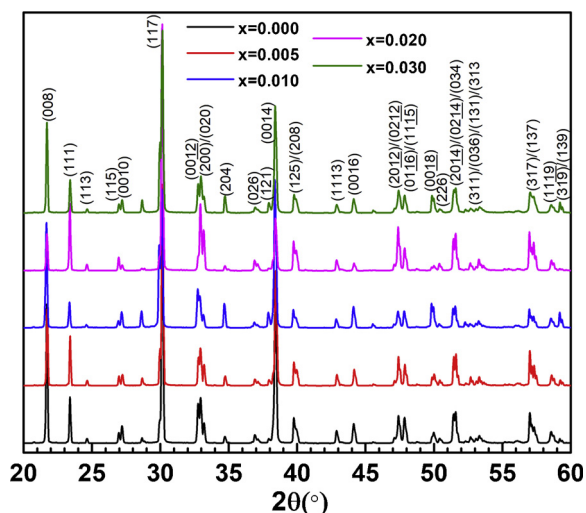


Fig. 1. XRD patterns of BTWN-100x ceramics.

PCB 9155D piezoelectric accelerometer calibration system in the range of 25–600 $^\circ\text{C}$.

3. Results and discussion

Fig. 1. Shows the XRD patterns of BTWN-100x ceramics. The results of XRD reveal that the main phase has a $\text{Bi}_4\text{Ti}_3\text{O}_{12}$ orthorhombic structure (JCPDS No. 72-1019), indicating that the dopants have been successfully incorporated into the B-site of the pseudo-perovskite layer as desired according to the principle of similar ionic radius (W^{6+} : 0.600 Å, Nb^{5+} : 0.640 Å, Ti^{4+} : 0.605 Å) [18].

To further elucidate the crystal structure of BTWN-100x, two compositions with $x = 0$ and 0.01 were chosen for XRD Rietveld refinements with the space group B2cb [19] by the TOPAS software at room temperature. Fig. 2 shows the refined profiles and SEM images of polished cross-sections after thermal etching for BIT and BTWN-1 ceramics. The plate-like grains having a large aspect ratio were observed for the BIT and BTWN-1 ceramics, which is a typical characteristic of Aurivillius ceramics because of the anisotropy of their crystal structure.

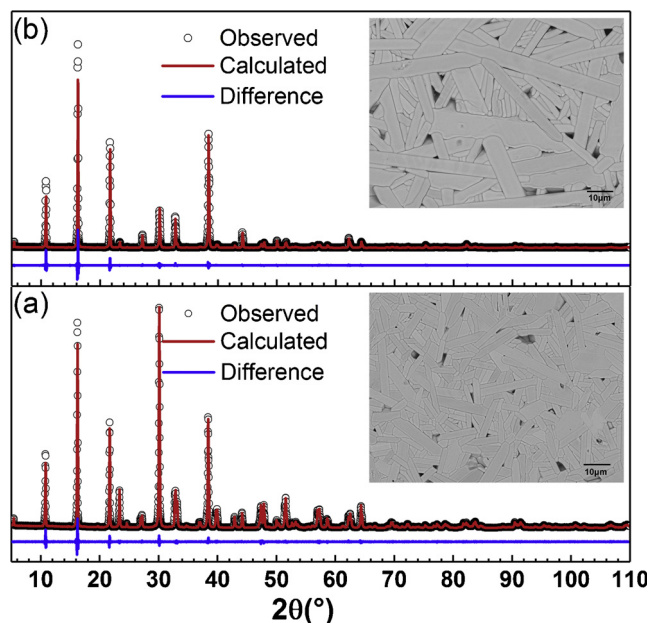


Fig. 2. XRD Rietveld refinements of (a) BIT and (b) BTWN-1, with an inset showing SEM images.

Table 1

Crystal data and lattice parameters for BIT and BTWN-1.

Compound	BIT	BTWN-1
Chemical formula	$\text{Bi}_4\text{Ti}_3\text{O}_{12}$	$\text{Bi}_4\text{Ti}_{2.98}\text{W}_{0.01}\text{Nb}_{0.01}\text{O}_{12}$
Crystal system	Orthorhombic	Orthorhombic
Space group	B2cb (41)	B2cb (41)
a(Å)	5.44970(15)	5.4499(3)
b(Å)	5.41051(15)	5.4112(3)
c(Å)	32.8218(8)	32.8192(13)
V(Å ³)	967.77(4)	967.86(9)
T(K)	298	298
R _{wp} (%)	5.17	9.51
R _p (%)	3.61	6.38
R _{exp} (%)	2.78	2.22
orthorhombic distortion(b/a)	0.9928	0.9929
tetragonal strain(c/a)	6.0227	6.0220

Some pores were observed at the intersecting corners of grains, which may be caused by the shedding of small grains during polishing.

The reliability factors R_{wp} , R_p and R_{exp} are 5.17%, 3.61% and 2.78% in pure BIT; and for BTWN-1 the values of the three factors are 9.51%, 6.38% and 2.22%, respectively. Table 1 lists crystal data and lattice parameters for BIT and BTWN-1 ceramics. The changes in structure of BLSFs ceramics can be determined by the structural distortion parameters [20,21], for example, orthorhombic distortion (b/a) and tetragonal strain (c/a). For pure BIT, the values of b/a and c/a are 0.9928 and 6.0227, respectively. Whereas, there is a slightly change in values of b/a (0.9929) and c/a (6.0220) for BTWN-1, indicating the substitution B-site by W/Nb can induces the variation of orthorhombic distortion and tetragonal strain of BIT-based ceramics. Moreover, the incorporation of W/Nb into BIT-based ceramics relieve the tetragonal strain, and in favor of improve the stability of the crystalline structure [22].

Fig. 3 shows Raman spectra of the BTWN-100x ceramics. The measured spectra are divided into 22 natural modes according to Lorentz fitting. It is well known that for BLSFs the low-frequency modes

(< 200 cm^{-1}) stem from the vibrations of the heavier Bi^{3+} ions, while the vibration modes of the BO_6 octahedra are above 200 cm^{-1} [23]. The modes at 62 cm^{-1} and 118 cm^{-1} are relevant to the vibration of the Bi^{3+} ions in the $(\text{Bi}_2\text{O}_2)^{2+}$ layers and the pseudo-perovskite units [24,25]. No shifts can be observed at these modes (62 cm^{-1} and 118 cm^{-1}) in BTWN-100x, indicating that W/Nb substitution has almost no influence on the $(\text{Bi}_2\text{O}_2)^{2+}$ layer and the A-site ions in the pseudo-perovskite.

The modes at 536 cm^{-1} (B_{2g}) and 562 cm^{-1} (B_{3g}) correspond to a relative displacement of apical oxygen atoms of the octahedra with opposite direction, and the mode at 614 cm^{-1} (A_{1g}) is related to the TiO_6 octahedron symmetric stretching mode [26]. The three modes mentioned above all present the shifts of high frequency, which implies that W/Nb was incorporated into the B-site of pseudo-perovskite structure and thereby induced distortion of the TiO_6 octahedron. Additionally, when the x increases from 0 to 0.030, the full width at half maximum (FWHM) of the three modes increases from 32 to 37, 30 to 34, and 36 to 54, respectively. The gradual broadening of modes 536 cm^{-1} , 562 cm^{-1} and 614 cm^{-1} may result from local lattice distortion of the TiO_6 octahedron, including degradation of symmetry of the local structure, the change in the lattice constant and the variation of phonon vibration frequency [27]. The results are consistent well with the refined XRD profiles discussed previously.

Fig. 4 shows the temperature dependence of the relative permittivity ϵ_r and dielectric loss $\tan\delta$ at 1 MHz. The T_c of pure BIT is 665 °C, which is consistent with the temperatures reported in the literature [28]. The T_c of BTWN-100x ceramics slightly decreases with the W/Nb modification and the T_c decreases to 640 °C for x = 0.030. In addition, the value of $\tan\delta$ of W/Nb-modified BIT ceramics decreases significantly and it is lower than that of pure BIT ceramic by approximately two orders of magnitude in the range of 25–600 °C. This may be due to the reduction of oxygen vacancies caused by W/Nb doping, which is conducive to the increment of activation energy, and resulting in $\tan\delta$ decrease [29]. Obviously, introducing donor additives of W/Nb at the B site of BIT can reduce oxygen vacancies effectively, as clarified

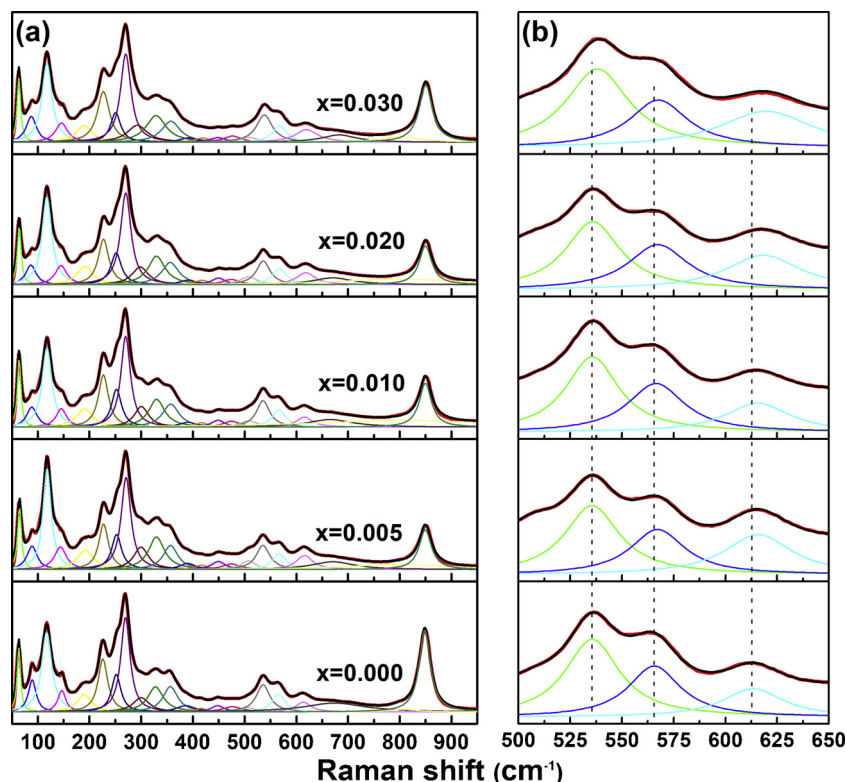


Fig. 3. Raman spectra of BTWN-100x ceramics at room temperature.

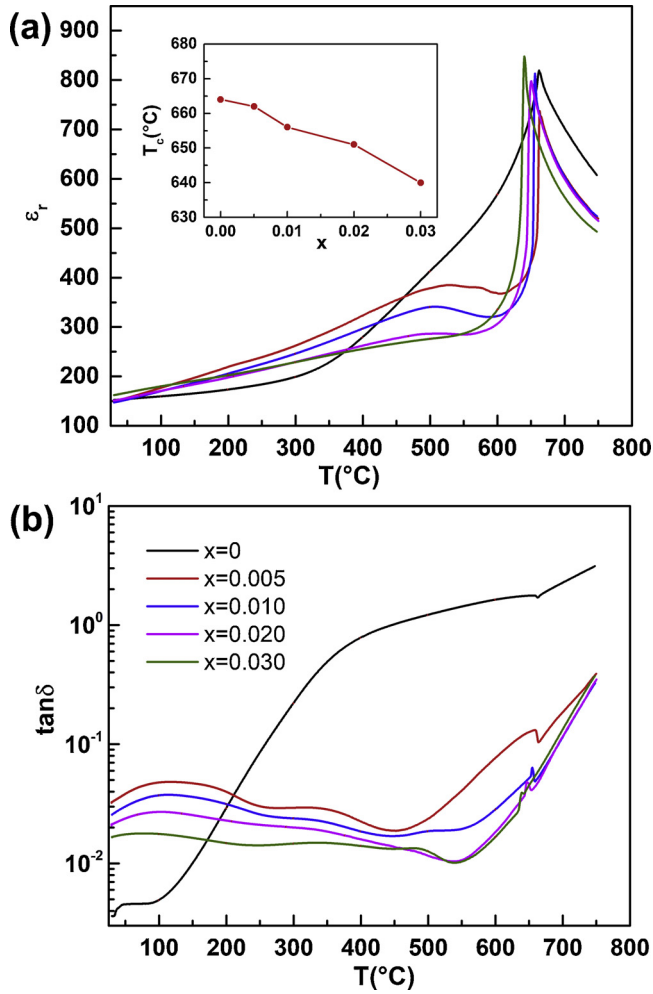


Fig. 4. The temperature dependence of the permittivity ϵ_r and dielectric loss $\tan\delta$ for BTWN-100x.

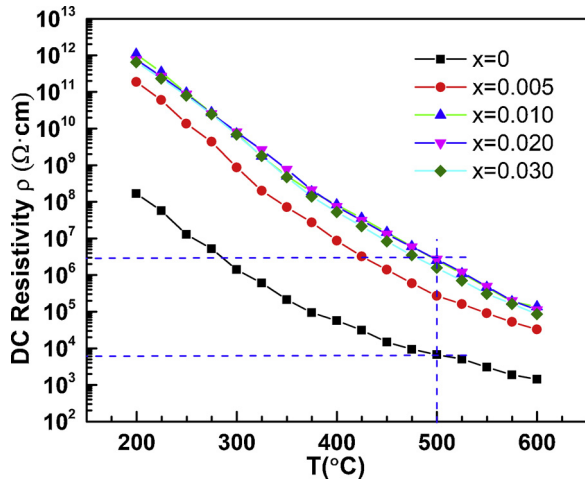


Fig. 5. Temperature dependence of DC resistivity of BTWN-100x ceramics.

by the defect reactions:

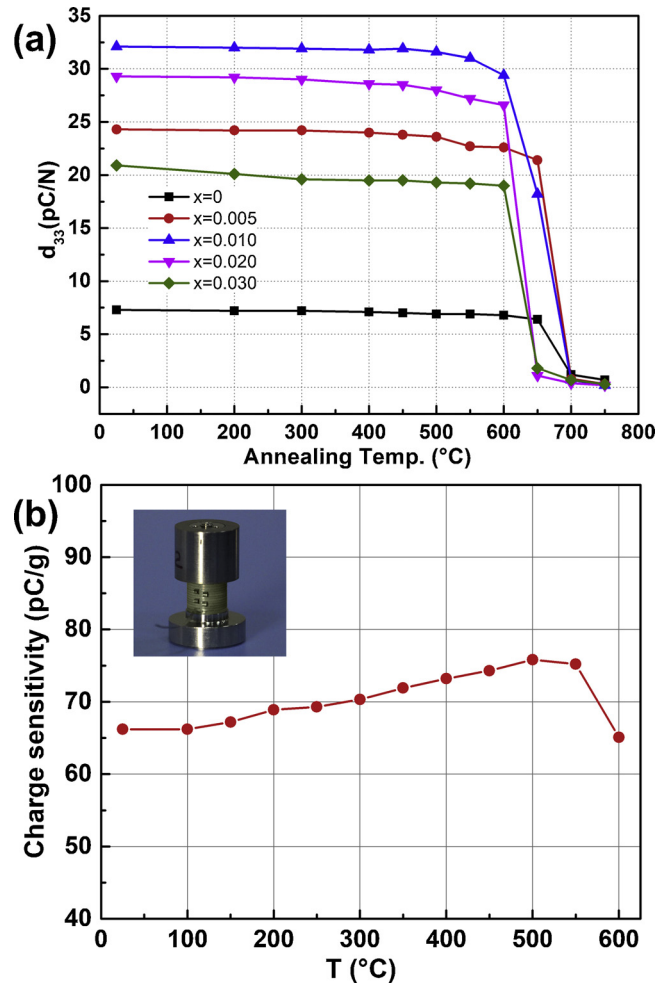


Fig. 6. (a) Temperature characteristic of piezoelectric coefficient d_{33} of BTWN-100x ceramics, (b) charge sensitivity of compression-mode accelerometer as a function of temperature.

Therefore, B-site doping with high valence ions can significantly reduce oxygen vacancies so that W/Nb-modified BIT samples can be effectively polarized and their piezoelectric properties enhanced. Additionally, the concentration of holes ($h^\bullet$) can be decreased through donor doping, resulting in an improvement of the resistivity.

The temperature dependence of the DC resistivity of all samples is shown in Fig. 5. The pure BIT shows a relatively low resistivity in the temperature range of 200–600 °C. The DC resistivity of BIT can be improved significantly by W/Nb addition. The DC resistivity for the x = 0.010 sample is $3 \times 10^6 \Omega \cdot \text{cm}$ at 500 °C, which is approximately three orders of magnitude higher than that of pure BIT. It is confirmed that the nominal valence state of Bi in the BIT ceramic lower than 3+, generating defect holes and leading to p-type conduction [11,30]. When Ti^{4+} was substituted by W^{6+} and Nb^{5+} , extra positive charge centers occur in the Ti-sites and extra electrons are generated according to the electro-neutrality principle. The neutralizing effect between electrons and holes decreases the concentration of carriers, which results in an increase of the resistivity. The generation of holes and neutralizing reaction as follows:



High resistivity allows for the application of high electric fields during poling at higher temperature, and at this point high piezoelectric activity can be achieved. Furthermore, the high resistivity at high

Table 2
Electrical properties of B-site modified BIT ceramics.

Materials	T_c (°C)	d_{33} (pC/N)	$\rho(\Omega\text{-cm})@ 500^\circ\text{C}$	Ref.
$\text{Bi}_4\text{Ti}_{3-2x}\text{Nb}_x\text{Ta}_x\text{O}_{12}$ ($x = 0.01$)	~668	26	/	[12]
$\text{Bi}_4\text{Ti}_{3-x}\text{W}_x\text{O}_{12} + 0.2 \text{ wt. \% NiO}$ ($x = 0.12$)	624	~18.5	5×10^6	[13]
$\text{Bi}_4\text{Ti}_{2.95}\text{W}_{0.05}\text{O}_{12.05} + 0.2 \text{ wt. \% Cr}_2\text{O}_3$	640	28	/	[14]
$\text{Bi}_4\text{Ti}_{2.9}\text{W}_{0.1}\text{O}_{12} + 4 \text{ at\% MnO}_2$	655	21	6×10^6	[15]
$\text{Bi}_4\text{Ti}_{3-2x}\text{W}_x\text{Nb}_x\text{O}_{12}$ ($x = 0.01$)	655	32	3×10^6	This work

temperatures can reduce charge drift and is also beneficial to avoiding coupling with the piezoelectric induced charges and then the practical application of piezoelectric ceramics in high-temperature conditions [31].

Fig. 6(a) shows the temperature characteristic of the piezoelectric activities for all samples after thermal annealing treatment. It is observed that the d_{33} values of the ceramics have a good temperature stability when the annealing temperature is below 600°C . Significantly, the BTWN-1 possesses a larger d_{33} of 29.4 pC/N with an annealing temperature of up to 600°C . This indicates that the piezoelectric property of BTWN-1 has a very good temperature stability. The d_{33} of BTWN-100x ceramics decreases rapidly as the annealing temperature approaches to T_c , and the values of d_{33} are almost zero when the temperature is above T_c . The electrical properties of the B-site modified BIT ceramics are listed in the Table 2. A large piezoelectricity, excellent temperature stability and high resistivity of W/Nb-modified BIT ceramics have been obtained in our work. This result could be ascribed to the increasement of TiO_6 octahedral distortion and the reduction of oxygen vacancies.

To further understand the high temperature performance of W/Nb-modified BIT-based ceramics, the *in-situ* temperature stability of the compression-mode sensor was evaluated by testing the charge sensitivity. Fig. 6(b) shows the charge sensitivity of the sensor at 160 Hz (1000 rad/s) in the range of 25– 600°C . The sensor is composed of three parts: the piezoelectric element (eleven layers of BTWN-1 ceramics with parallel connection), seismic mass block and substrate. The piezoelectric element is between the seismic mass and the substrate and is fastened by a screw. The charge sensitivity of the sensor is 70 ± 6 pC/g, with little variation in the measure temperature range of 25– 600°C . The excellent temperature stability of W/Nb-modified BIT ceramics and the good *in-situ* temperature stability of compression-mode sensor, implying that W/Nb-modified BIT ceramics is a promising candidate for application at high temperatures of up to 600°C .

4. Conclusions

The high Curie temperature $\text{Bi}_4\text{Ti}_{3-2x}\text{W}_x\text{Nb}_x\text{O}_{12}$ (BTWN-100x, $x = 0.00, 0.005, 0.010, 0.020$ and 0.030) Aurivillius piezoelectric ceramics were investigated in detail. The results of XRD refinements and Raman spectroscopy demonstrate that W/Nb substitution for the Ti-site of the pseudo-perovskite results in the enhancement of structure distortion, as well as in the reduction of tetragonal strain and the symmetry of the local lattice structure. The T_c of BTWN-100x decreased slightly with increasing W/Nb substitution with a T_c are in the range of 640 – 665°C . The BTWN-1 ceramic has a high Curie temperature (T_c) of 655°C and an optimum piezoelectric coefficient (d_{33}) of 32 pC/N . The piezoelectric activity of BTWN-1 ceramic shows stability against temperature, and a higher d_{33} of 29.4 pC/N is obtained when the annealing temperature is up to 600°C . The *in-situ* temperature stability of a compression-mode sensor in the range of 25– 600°C is comparatively stable. These results reveal that the W/Nb-modified BIT piezoelectric ceramics are excellent candidates for applications in the high-temperature field.

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